

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

Student Name: A.Rohit

Roll No.: A24126552067

Date: 08-09-2026

1. TITLE

JavaScript Basics - Arrays and Functions

2. AIM

To understand and implement JavaScript basics including variables, array iteration, and function declarations.

3. OBJECTIVE

The objectives of this experiment are:

- Initialize and iterate through JavaScript Arrays.
 - Create reusable functions.
 - Use DOM manipulation to inject calculated values.
-

4. THEORY

JavaScript allows dynamic behavior on web pages. Arrays hold multiple values, and functions encapsulate logic for reuse. The DOM API allows JS to modify the HTML document dynamically.

5. ALGORITHM / PROCEDURE

1. Create a basic HTML structure with an output container.
 2. Initialize arrays in a script block.
 3. Write functions to process array data.
 4. Inject the processed output into the DOM container.
-

6. PROGRAM

Source File: week3rd.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>JS Basics - Arrays & Functions</title>
</head>
<body>
  <h1>JavaScript Arrays & Functions Demonstration</h1>
  <div id="output"></div>

  <script>
    // Array Initialization
    const marks = [85, 92, 78, 64, 95, 88];

    // Function 1: Calculate Average
    function calculateAverage(arr) {
      const sum = arr.reduce((acc, curr) => acc + curr, 0);
      return (sum / arr.length).toFixed(2);
    }

    // Function 2: Filter High Scores (Marks >= 80)
    const filterHighScores = (arr) => arr.filter(score => score >= 80);

    // Function 3: Find Max and Min Marks
    function getMinMax(arr) {
      return {
        max: Math.max(...arr),
        min: Math.min(...arr)
      };
    }

    // Execution
    const avg = calculateAverage(marks);
    const topMarks = filterHighScores(marks);
    const { max, min } = getMinMax(marks);

    // Output to Browser Console
    console.log("Original Marks:", marks);
    console.log("Average Marks:", avg);
    console.log("Top Marks (>= 80):", topMarks);
    console.log(`Highest Mark: ${max}, Lowest Mark: ${min}`);

    // Output to Web Page DOM
    document.getElementById("output").innerHTML = `
      <p><strong>Original Marks:</strong> ${marks.join(", ")}</p>
      <p><strong>Average Marks:</strong> ${avg}</p>
      <p><strong>Top Scores (&ge; 80):</strong> ${topMarks.join(", ")}</p>
      <p><strong>Highest:</strong> ${max} | <strong>Lowest:</strong> ${min}</p>
    `;
  </script>
</body>
</html>

```

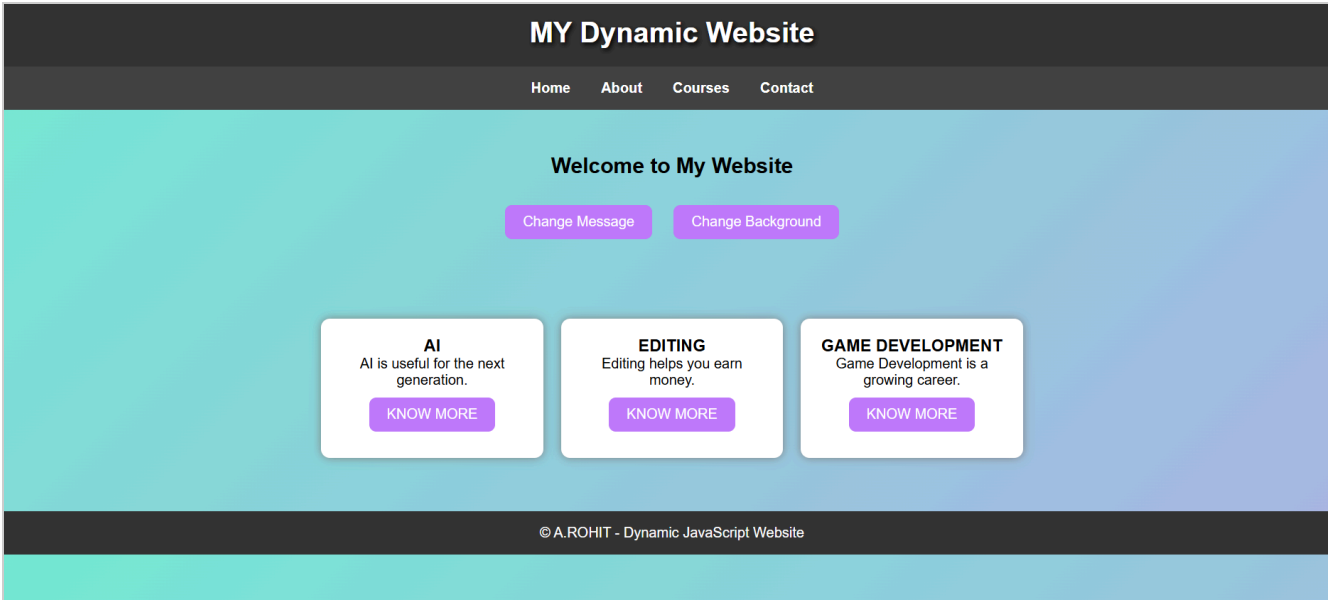
7. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Action	Expected Result	Status
TC-1	Execute JS script	Array data is correctly parsed and displayed in DOM	Pass

8. OUTPUT

Output 1



9. RESULT

Thus, JavaScript basics including arrays and functions were successfully practiced and verified.